

DHRUV NIRMAL

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EXECUTIVE SUMMARY

Data Engineer with 2+ years of experience building scalable data pipelines and analytics systems across **5 enterprise datasets**. Designed and deployed **20+ automated ETL pipelines** processing millions of transactions monthly, reducing manual reporting effort by 40% and improving reporting turnaround by 50%. Experienced in **Python, SQL, and workflow automation** with hands-on experience building **machine learning models, AI-driven analytics workflows**, and automated reporting systems to support data-driven decision making.

KEY SKILLS

Programming & Data Engineering

Python, SQL, Snowflake, Data Modelling, Data Cleaning, Feature Engineering

Cloud & Infrastructure

Azure, AWS (S3, Lambda, IAM, EC2), Databricks, Docker, Terraform (Infrastructure-as-Code)

Machine Learning & AI

Scikit-learn, TF-IDF Vectorisation, Logistic Regression, Forecasting Models, LLM Workflows (Claude, ChatGPT)

Visualisation & BI

Power BI, Tableau, Data Storytelling, KPI Analytics

EXPERIENCE

Purchasing Index Data Analytics, [Data Engineer](#)

June 24 – Present

Deliver data analytics and reporting solutions in a client-facing consulting role, supporting large-scale supplier and transaction datasets for enterprise clients across Australia and New Zealand. Translate complex datasets into actionable insights through automated reporting pipelines, dashboard development, and statistical analysis to support business decision-making.

- Built and maintained 20+ automated ETL pipelines processing millions of procurement transactions monthly across **5 enterprise clients**, using Python, SQL, and batch files, deployed on Azure-hosted servers with Windows Task Scheduler to deliver reliable, production-grade analytics pipelines.
- Automated client reporting and analytics workflows using Python, SQL, and KNIME, integrating LLM tools such as **Claude** and ChatGPT to accelerate development of analytics scripts and streamline delivery of reporting
- Developed and maintained **16+ Tableau dashboards** enabling procurement and finance teams to monitor supplier spend, track KPIs, and identify cost optimisation opportunities.
- Performed anomaly detection on procurement datasets, identifying cost leakage patterns that contributed to ~15% client savings.
- Built and deployed client-facing analytics platforms and dashboards across 5 enterprise clients, communicating complex technical outputs as clear, actionable insights for senior executives and business stakeholders.

Machine Learning & AI

- Built a machine learning model to predict chemical consumption for industrial clients, enabling improved production planning and inventory management.
- Developed a binary classification model using TF-IDF vectorisation and Logistic Regression to categorise transaction descriptions to cut manual effort by 40% and improve data standardisation for downstream analytics.
- Built an agentic AI workflow using Claude to automatically generate structured client progress reports from analytics outputs.
- Developed a multi-agent data QA validation workflow with five agents and an orchestrator to validate client datasets prior to delivery.
- Implemented an LLM-driven sentiment analysis workflow to analyse sales call transcripts and identify customer sentiment trends.

- Built an automated presentation generation workflow using Claude to convert analytics outputs into PowerPoint decks.

Victorian Centre for Data Insights (VCDI), [Data Engineer Intern](#)

Aug 23 – Nov 23

Designed distributed data processing pipelines and machine learning workflows for government procurement monitoring.

- Built a distributed anomaly detection pipeline in Databricks using PySpark, improving detection accuracy by 20%
- Engineered scalable transformation layers to process procurement data in cloud-based environments
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders
- Presented system architecture and model outcomes to cross-functional stakeholders, bridging technical and business contexts

Monash University, [Master in Data Science](#)

Feb 22 – Dec 23

- Courses: Statistics I and II, Machine Learning, Communicating with Data, Supply Chain Management, High Dimensional Analysis, Web App Development, Applied Forecasting.

Terminal Ballistics Research Laboratory (TBRL), [Research Data Analyst](#)

Jan 21 – Jul 21

Worked with Head of Blast and Damage Studies to acquire, analyse, and filter noise using butterworth filter from blast-wave data, resulting in a 20% reduction in noise levels, making better defence equipment for the Defence Research and Development Organisation.

- Data Analysis: Analysed blast wave data using advanced statistical models to predict noise levels, enhancing the design efficiency of defence equipment with a 20% increase in prediction accuracy.
- Report Writing: Compiled and synthesised blast wave analysis findings into a comprehensive 70-page report, utilising visualisation tools like MATLAB and Excel, thereby aiding the TBRL team in refining defence equipment designs by 10%.

Thapar University, [Bachelor in Engineering](#)

May 17 –May 21

- Courses: Statistics, C, C++ , Analog and Digital Electronics, Entrepreneurship, Mathematics I, II and III.

PROJECTS

Cloud Data Platform with Terraform | AWS, Terraform, Snowflake, Docker, PowerBI

- Designed and provisioned complete cloud infrastructure using Terraform, including S3 buckets, IAM roles, EC2 compute, and networking components
- Implemented Infrastructure-as-Code to enable reproducible, version-controlled environment deployment
- Built automated data ingestion pipeline from external APIs to Snowflake deployed on Terraform-managed infrastructure
- Applied modular Terraform design patterns to support scalable multi-environment deployment (dev, staging, prod)